

Supply Chain Management (SCM) Business Architecture

Overview

This document defines the Supply Chain Management (SCM) module for the Zeus ERP system. The module governs the acquisition of raw components from hardware vendors to feed internal production lines and subsequently manages the distribution of finished hardware to clients.

1. Core Subsystems

Vendor Routing

Map specific raw materials to approved suppliers to automate procurement. This subsystem provides the sourcing intelligence required to convert an MRP deficit into an actionable purchase order.

Core Data Structures:

- **Supplier Profile:** Maintains the master record of registered vendors. Tracks active status and holds dynamically calculated performance scores.
- **Vendor-SKU Mapping:** A junction entity linking an internal raw material SKU to a specific supplier ID. Stores operational parameters for the transaction: unit price, minimum order quantity, and lead time in days.

Resolution Logic: When the system receives a component shortage alert, it executes a lookup against the Vendor-SKU mapping. It filters for active suppliers matching the required SKU and sorts them dynamically based on calculated performance metrics and unit pricing. The engine extracts the optimal price and lead time to construct the draft PO payload. Operators can override the system-recommended vendor to address emergencies or operational constraints; deviations from the optimal routing are warned in the audit log for administrative review.

Metrics Calculation: Supplier performance is evaluated dynamically to ensure routing logic relies on current data. The engine queries historical Goods Receipt records to calculate:

- **On-Time Rate:** Percentage of orders received on or before the expected delivery date.
- **Defect Rate:** Ratio of defective component quantities against total received quantities.

Purchase Order (PO) Orchestration

Consolidate raw material shortages from the MRP engine into vendor-specific purchase orders. Translate quantitative deficits into executed financial documents.

Execution Mechanics:

- **Deficit Aggregation:** The system aggregates discrete MRP shortage notifications into a global deficit pool per SKU. Operators purchase against this combined pool to leverage

bulk pricing and meet minimum order quantities, while the system tracks the underlying allocation to specific production quotas.

- **Mono-Vendor Constraint:** Drafts are strictly mono-vendor. Each PO represents a distinct legal contract with a single supplier. Resolving a mixed-vendor deficit requires the operator to execute multiple, separate drafts.
- **Concurrency Control (Eager Slot-Locking):** Prevent duplicate procurement entries across multiple operators utilizing a step-by-step eager lock mechanism:
 1. An operator initiates a draft and selects a target supplier.
 2. The operator selects a component SKU and enters the desired purchase quantity.
 3. The system immediately deducts this quantity from the global deficit pool and checks the status atomically. All parallel reads and writes to this deficit pool are queued.
 4. If the entered quantity exceeds the current deficit pool, the system issues a warning. The operator may proceed (purchasing excess stock), but the alert flags the overage.
 5. The quantity is slot-locked. The operator continues to add additional items or complete review steps.
 6. The slot-lock persists for exactly 30 minutes. If the operator finalizes the draft, the lock becomes permanent. If the 30-minute window expires before finalization, the draft is voided, and the locked quantities revert to the open deficit pool.
 7. The system logs a warning anomaly if a draft voids due to operator inactivity.
- **State Management:** Enforce a strict, one-way state machine for PO lifecycles:
 - **Draft (Pending):** Initial state upon creation. Fully editable until finalized or voided by timeout.
 - **Approved (Sent):** Locked for editing. Represents a committed financial obligation dispatched to the vendor. Requires manual authorization.
 - **In Transit:** Vendor confirms shipment. Provides the warehouse with expected delivery dates and inbound freight manifests.
 - **Received / Partial:** Triggered exclusively by the Goods Receipt subsystem. Closes the PO line items based on physically verified stock.
- **State Regression Prevention:** Block any attempts to revert terminal or active states (e.g., reverting a Received or In Transit order back to Draft) to maintain data integrity across the system.
- **Bulk Operations:** Execute batch state transitions (e.g., bulk approving multiple Draft POs) via atomic operations. The system validates each PO's current state before applying the transition, skipping invalid entities to prevent workflow halts.

Goods Receipt

Act as the physical validation and ingestion point for inbound shipments. Map all warehouse dock receipts strictly to existing Purchase Orders.

Execution Mechanics:

- **State Progression:** Advance Goods Receipts through a linear state machine: Pending (awaiting arrival), Inspected (quantities verified), Complete (finalized and written to ledger), or Discrepancy (flagged for review).
- **Concurrency Control (Parallel Lock Procedure):** Maintain manifest availability while preventing conflicting write operations:
 - **View-Only Access:** Secondary operators may open a manifest normally at any time. The "Admit/Receive" button remains active unless a lock is explicitly propagated.
 - **Lock Propagation:** A lock is only requested when an operator clicks the "Admit" or "Start Inspection" button.
 - **Parallel Conflict Resolution:**
 - If a lock has already been propagated by another operator, the "Admit" button is immediately greyed out for all other users.
 - If two operators click the button simultaneously, the first successful request propagates the lock. All subsequent parallel requests return an immediate error alert to the operator and grey out the button.
 - A notification banner identifies the primary operator holding the lock to facilitate coordination.
 - **Lock Expiration:** The lock remains active for 60 minutes of inactivity. If the primary operator disconnects or fails to submit progress, the lock is automatically released, re-enabling the "Admit" button for other operators.
- **Software-Level Quantity Verification:** Implement "blind receiving" to ensure physical counts are conducted. Force operators to manually type the exact physical count of received and defective units into empty inputs. The system prohibits "Receive All" or pre-filled quantity buttons to ensure a best-effort software-level validation of the physical stock.
- **Automated Defect Quarantine (Age & Stability Logic):**
 - **Aging Thresholds:** For components prone to storage degradation (e.g., lithium-ion batteries), the system forces the entry of a production date. If the current date exceeds the predefined stability threshold (e.g., 5-year-old laptop batteries), the system automatically flags the units as defective and logs a warning for audit.
 - **Stability Exemption:** For stable components (e.g., RAM sticks), the system bypasses chronological age checks, provided there is no evidence of corrosion, or the component is not phased out by the industry.
- **Discrepancy Origin & Audit Tracing:**
 - **Inbound Discrepancy:** Triggered at the dock if physical counts do not match the PO manifest or if items are rejected due to aging logic.
 - **Downstream Traceability:** If a number discrepancy or SKU mismatch is detected during production, the system utilizes the component batch/lot ID to trace liability back

to the specific dock entry.

- **Atomic Ledger Update:** Trigger a transactional database sequence upon state change to Complete:
 1. Write an Inbound transaction to the Inventory Ledger Entry table.
 2. Update the Inventory Snapshot to reflect the new available balance based strictly on the verified received quantity.
 3. Cascade a status update to the parent PO: set to Received if all line items are fully fulfilled, or Partial if outstanding quantities remain.

Downstream Logistics

Manage the outbound supply chain. Track the dispatch of finished, assembled hardware from the warehouse to distribution centers, enterprise clients, or end consumers.

Execution Mechanics:

- **Dispatch Orchestration:** Ingest finalized sales orders from the Sales module. Group outbound shipments by destination to optimize carrier selection and generate warehouse pick lists.
- **Concurrency Control (Dispatch-Locking Procedure):** Prevent duplicate shipments or conflicting tracking assignments for the same outbound order:
 - **View-Only Access:** All dispatchers can view pending outbound manifests. The "Pack Order" and "Dispatch" buttons remain active unless a lock is active.
 - **Lock Propagation:** Trigger an atomic session lock when an operator clicks "Pack Order" to begin physical assembly of the shipment.
 - **Parallel Conflict Resolution:** If secondary operators attempt to pack the locked order, the action buttons are greyed out, and a banner displays the primary operator's ID. Simultaneous lock attempts are resolved atomically, returning an error to the rejected requests.
 - **Lock Expiration:** The lock persists for 30 minutes of inactivity to accommodate physical packing times. Expiration releases the lock back to the global queue. Finalizing the dispatch permanently clears the lock.
- **Outbound State Management:** Track the shipment lifecycle through a linear state machine: Packing (inventory allocated and locked), Dispatched (physically handed to the carrier), and Delivered (confirmed receipt by the client).
- **Freight Data Capture:** Log exact shipping costs, carrier IDs, and tracking numbers per outbound order. Feed this data to the Sales and Financial modules to calculate final delivery margins and provide client-facing tracking visibility.
- **Inventory Deduction Trigger:** Upon transitioning an order to the Dispatched state, trigger an Outbound transaction to the Product module's Inventory Ledger to decrement the finished goods balance, synchronizing system stock with physical warehouse depletion.

2. System Interconnections

The SCM module functions as the central router between demand generation, physical storage, and financial reporting. Interconnections are delineated by subsystem and external dependency.

Vendor Routing

Data Inputs:

- **From the MRP:** Quantitative component shortages (SKU, quantity, and required date) to trigger vendor selection.
- **From Product/BOM Catalog:** Active SKU definitions and hardware specifications to validate component compatibility during mapping.

Data Outputs:

- **To PO Orchestration:** Selected vendor ID, current unit price, and projected lead time to populate draft purchase orders.

Purchase Order Orchestration

Data Inputs:

- **From Vendor Routing:** Commercial vendor metadata and pricing to initialize order records.
- **From the MRP:** Target build identifiers (order IDs) to link PO line items to specific manufacturing requirements for audit trails.

Data Outputs:

- **To Goods Receipt:** Finalized PO manifests (SKUs and ordered quantities) to serve as the baseline for inbound warehouse validation.
- **To Financial Module:** Total order value and expected payment terms upon PO approval to track upcoming capital expenditures.

Goods Receipt

Data Inputs:

- **From PO Orchestration:** Active "In Transit" manifests to authorize physical dock reception.

Data Outputs:

- **To Product/Inventory Ledger:** Physically verified inbound quantities and storage locations to increase active inventory stock.
- **To PO Orchestration:** Actual received vs. defective counts to trigger PO status transitions (Received or Partial).
- **To Vendor Routing:** On-time performance and defect data to dynamically update supplier quality scores.

Downstream Logistics

Data Inputs:

- **From Sales:** Finalized sales orders, shipping priorities, and customer destination addresses to initialize dispatch workflows.

Data Outputs:

- **To Product/Inventory Ledger:** Decrement requests for finished goods upon shipment dispatch.
- **To Sales:** Real-time tracking IDs and carrier statuses to update customer order dashboards.